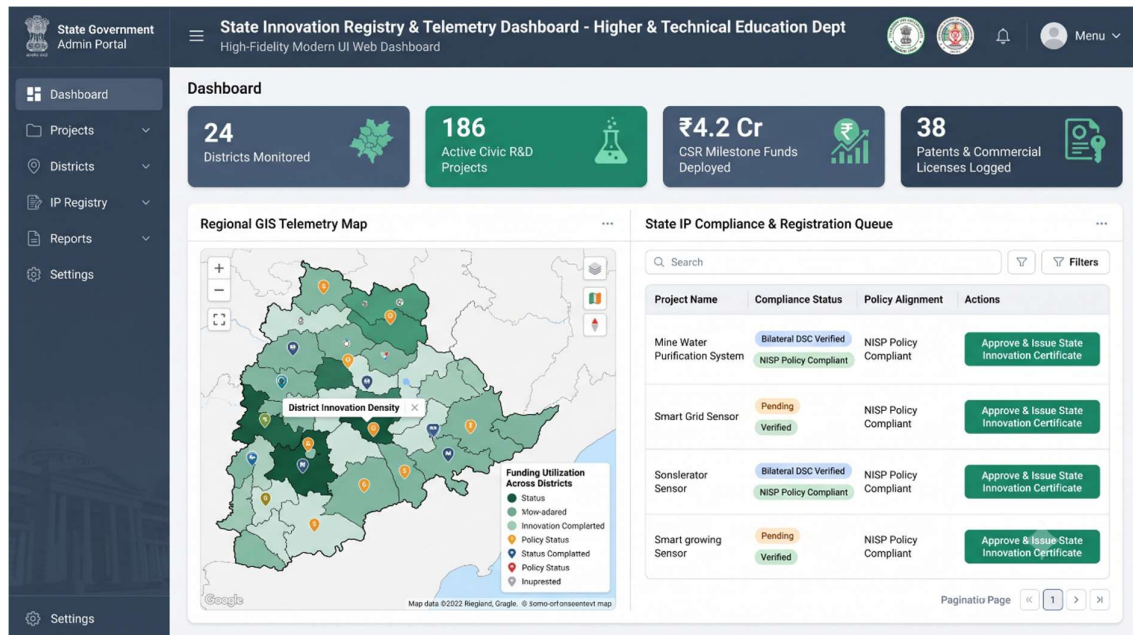




1. Statewide Telemetry & KPI Cards (Top Bar)

- **Districts Monitored:** Tracks innovation activity across all districts in the state.
- **Active Civic R&D Projects:** Shows ongoing university lab projects tackling real municipal challenges.
- **CSR Funds Deployed:** Financial metrics showing corporate capital moving through milestone escrow accounts directly into academic research.
- **Patents & Commercial Licenses Logged:** Verifiable count of intellectual property created within the state ecosystem.

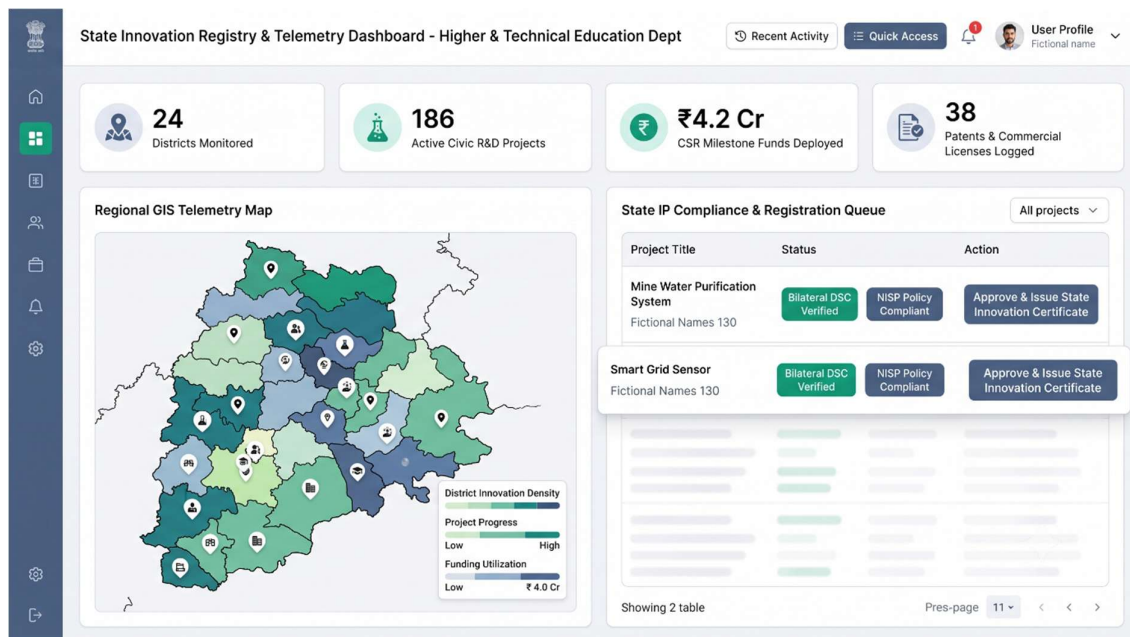
2. District Heatmap & Innovation Density (Left Side)

- A regional GIS telemetry view displaying real-time project distribution, active university hubs, and problem severity across administrative districts.
- Allows state supervisors to detect regional disparities (e.g., ensuring R&D resources reach remote districts, not just major urban centers).

3. State IP Compliance & Registration Queue (Right Side)

- **Regulatory Compliance Audit:** Automatic verification that bilateral agreements between universities and corporate sponsors adhere to **National Innovation and Startup Policy (NISP)** guidelines.
- **Digital Verification Badges:** Flags completed DSC (Digital Signature Certificate) sign-offs from both the University TTO Registrar and the Corporate Legal Officer.

- **Action Gate:** The state officer clicks "**Approve & Register**" to archive the agreement into the official State Innovation Registry.
- **Certificate Issuance:** Triggers the auto-generation of the tamper-evident **State Government Innovation Certificate** pushed directly to student/faculty DigiLocker credentials upon successful municipal field deployment.



Dashboard (Top Active Tab):

- **What it does:** The executive home screen showing the statewide summary KPIs, the GIS telemetry map, and the pending IP approval queue.

Projects:

- **What it does:** Displays all 186 active civic R&D projects across all 24 districts in Jharkhand.
- **When clicked:** Opens a master table where officers can search and filter builds by university, lead corporate sponsor, technology domain (e.g., IoT, Clean Energy, Water Treatment), and current Technology Readiness Level (TRL).

Districts:

- **What it does:** Management portal for all 24 District Nodal Officers (DNOs).
- **When clicked:** Displays performance scorecards for each district: total citizen complaints logged, triage resolution speeds, and the percentage of local issues routed to municipal desks vs. academic engineering labs.

? IP Registry:

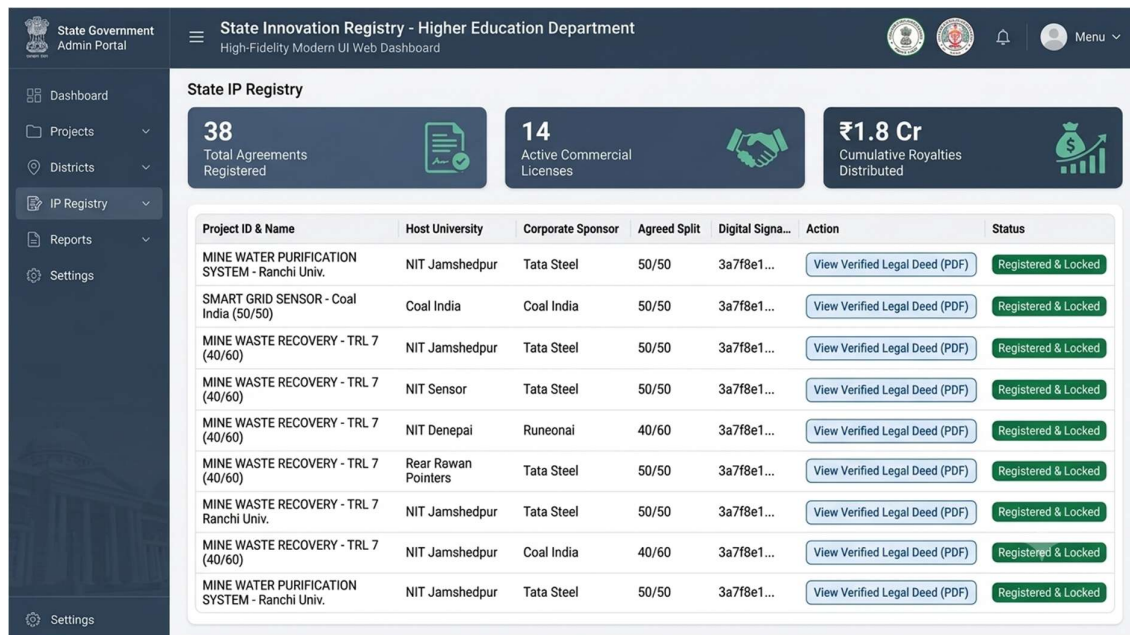
- **What it does:** The official state repository of signed legal agreements and patent filings.
- **When clicked:** Opens the legal audit ledger showing every approved Tripartite Agreement, digital signature hashes, and royalty distribution records (mockup generated below).

? Reports:

- **What it does:** The official compliance and export engine.
- **When clicked:** Allows state administrators to generate automated CSV/PDF reports for the Chief Minister's office, the Department of Higher Education, or central government bodies (like NITI Aayog / DST) tracking CSR fund utilization and patent yields.

? Settings & Log Out (Bottom Icons):

- **What it does:** Admin-level permission management (granting nodal credentials to new district officers) and secure session sign-out.



Project ID & Name	Host University	Corporate Sponsor	Agreed Split	Digital Signa...	Action	Status
MINE WATER PURIFICATION SYSTEM - Ranchi Univ.	NIT Jamshedpur	Tata Steel	50/50	3a7f8e1...	View Verified Legal Deed (PDF)	Registered & Locked
SMART GRID SENSOR - Coal India (50/50)	Coal India	Coal India	50/50	3a7f8e1...	View Verified Legal Deed (PDF)	Registered & Locked
MINE WASTE RECOVERY - TRL 7 (40/60)	NIT Jamshedpur	Tata Steel	50/50	3a7f8e1...	View Verified Legal Deed (PDF)	Registered & Locked
MINE WASTE RECOVERY - TRL 7 (40/60)	NIT Sensor	Tata Steel	50/50	3a7f8e1...	View Verified Legal Deed (PDF)	Registered & Locked
MINE WASTE RECOVERY - TRL 7 (40/60)	NIT Denepai	Runeonai	40/60	3a7f8e1...	View Verified Legal Deed (PDF)	Registered & Locked
MINE WASTE RECOVERY - TRL 7 (40/60)	Rear Rawan Pointers	Tata Steel	50/50	3a7f8e1...	View Verified Legal Deed (PDF)	Registered & Locked
MINE WASTE RECOVERY - TRL 7 Ranchi Univ.	NIT Jamshedpur	Tata Steel	50/50	3a7f8e1...	View Verified Legal Deed (PDF)	Registered & Locked
MINE WASTE RECOVERY - TRL 7 (40/60)	NIT Jamshedpur	Coal India	40/60	3a7f8e1...	View Verified Legal Deed (PDF)	Registered & Locked
MINE WATER PURIFICATION SYSTEM - Ranchi Univ.	NIT Jamshedpur	Tata Steel	50/50	3a7f8e1...	View Verified Legal Deed (PDF)	Registered & Locked

? **Cumulative Royalty & License Counters:** Tracks total registered agreements (38), active commercial licenses (14), and cumulative royalties disbursed back to host institutions (₹1.8 Cr).

? **Project & Partner Columns:** Clearly links the project ID to the host academic institution (e.g., BIT Mesra, NIT Jamshedpur) and corporate sponsor (e.g., Tata Steel, Coal India).

🔒 **Tamper-Evident SHA Hash & PDF Deed:** Displays the cryptographic hash of the document and includes a **[View Verified Legal Deed]** button that lets the government download the signed tripartite deed in case of future legal disputes.



- ## 2. Map Layer & Visualization Controls (Top-Right Corner)

-  **Stacked Layers Icon (Map Layer Switcher):**
 - **What it does:** Toggles between different data visualization overlays:
 - *Layer 1: Innovation Density (Default)* — Shaded by number of active college prototypes.
 - *Layer 2: CSR Fund Deployment* — Shows financial flow heatmaps (which districts are receiving the most corporate investment).
 - *Layer 3: Civic Problem Influx* — Visualizes where citizens are logging the highest concentration of unsolved municipal complaints.
- **IN /**  **Department Flag Filter:**
 - **What it does:** Filters map markers by state departments (e.g., *Drinking Water & Sanitation, Road Construction, Energy & Power, Urban Development*).
-  **Compass / Orientation Dial:**
 - **What it does:** Resets the geographic map orientation to true north after manual rotation or 3D tilt adjustments.

3. Pin Markers & Icons Displayed on District Polygons

-  **Graduation Cap Pin (Academic Lab Hubs):**
 - **What it indicates:** An accredited engineering college or research institution (e.g., BIT Mesra, NIT Jamshedpur, IIT ISM Dhanbad) currently building a funded prototype.
 - **Click action:** Pops up an institution card showing: *Active Student/Faculty Teams, Lab Facilities Available, and Current Pipeline of Deployed PoCs.*
-  /  **Industrial Building Pin (Corporate Sponsors & Testing Sites):**
 - **What it indicates:** A corporate sponsor providing CSR funding, lab access, or industrial testing grounds (e.g., Tata Steel facilities in Jamshedpur, Coal India/BCCL sites in Dhanbad).
 - **Click action:** Shows corporate commitment details: *Total Pledged CSR Capital, Active Mentors Assigned, and Industry Testing Facilities Opened.*
-  **Flask Pin (Active Bench Trials / Lab Prototyping):**

- **What it indicates:** A project currently at Technology Readiness Level 4 or 5 undergoing active physical hardware or chemical testing inside a campus laboratory.
-  **Amber Alert Marker (Unclaimed Civic Distress Zones):**
 - **What it indicates:** A critical municipal problem logged by citizens and verified by the District Nodal Officer, but **not yet claimed** by any university research lab or industry sponsor.
 - **Click action:** Gives the state admin a one-click button: **[Allocate State Seed Grant]** to incentivize universities to claim the challenge.

4. Interactive Hover Cards & Legends (Center & Bottom)

- **The Hover Tooltip (e.g., "Ranchi: 34 Active Builds, ₹1.2 Cr CSR Deployed"):**
 - **What it does:** An instant telemetry readout triggered when the administrator hovers the mouse over any district boundary.
 - **Data shown:**
 1. *District Name* and nodal officer in charge.
 2. *Total Active Builds* currently in development.
 3. *Total CSR/Govt Funds* locked and moving through milestone escrow.
 4. *Triage Ratio* (% routine repairs solved vs. % deep-tech R&D underway).
- **Bottom-Right Legend & TRL Metric Bars:**
 - **Color Palette Indicator:** Clear visual guide matching map shades to R&D activity (Dark Green = 20+ builds, Medium Teal = 5–15 builds, Pale Mint = 1–5 builds, Amber = high civic backlog).
 - **Funding Utilization Gauge:** Live progress meter displaying total statewide budget utilization against annual targets.